

YOUNG SU KO

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EDUCATION

University of California, San Diego

Ph.D. Biochemistry and Molecular Biophysics (GPA: 4.0/4.0)

La Jolla, CA

Expected September 2027

Dissertation: AI-guided Design and Prediction of Protein Interactions

Pomona College

B.A. Chemistry, *magna cum laude* (GPA: 3.99/4.0)

Claremont, CA

May 2022

RESEARCH EXPERIENCE

Graduate Student Researcher

January 2023 – Present

University of California, San Diego (Advisor: Wei Wang)

- Designed *forge*, a sequence-based generative model for protein binder design using flow matching over protein language model representations, enabling structure-free protein design.
- Applied AI-driven sequence optimization to engineer an EGFR-binding protein with 67% stronger affinity than native EGF in the Adapticv Bio binder design competition.
- Developed TUnA, a Transformer-based model for protein-protein interaction prediction, achieving state-of-the-art AUROC (0.70) on a leakage-free benchmark.
- Analyzed representation transfer by benchmarking six protein language models across six datasets; proposed an embedding fusion framework to leverage multiple representations.

Undergraduate Researcher

September 2019 – May 2022

Pomona College (Advisor: Roberto Garza-Lopez)

- Performed virtual screening of SARS-CoV-2 main protease inhibitors using molecular docking and molecular dynamics simulations.

PREPRINTS AND PUBLICATIONS

Y.S. Ko*, W. Wang. *forge: sequence-based binder design with latent flow matching*. **Machine Learning in Structural Biology (MLSB)**, 2025.

K. Devkota, D. Shonai, J. Mao, **Y.S. Ko**, W. Wang, S. Soderling, R. Singh. *Miniaturizing, Modifying, and Magnifying Nature's Proteins with Raygun*. **bioRxiv**, 2025.

L. Zhang, H. Guo, L. Schaffer, **Y.S. Ko**, D. Singh, H. Rahmani, D. Grotjahn, E. Villa, M. Gilson, W. Wang, T. Ideker, E. Xing, P. Xie. *ProteinAligner: A Multi-modal Pretraining Framework for Protein Foundation Models*. **bioRxiv**, 2024.

Y.S. Ko*, J. Parkinson, W. Wang. *Scalable embedding fusion with protein language models: insights from benchmarking text-integrated representations*. **Briefings in Bioinformatics**, 2026.

J. Parkinson, R. Hard, **Y.S. Ko**, W. Wang. *RESP2: An uncertainty-aware multi-target multi-property optimization AI pipeline for antibody discovery*. **Advanced Science**, 2025.

Y.S. Ko*, J. Parkinson, C. Liu, W. Wang. *TUnA: an uncertainty-aware transformer model for sequence-based protein-protein interaction prediction*. **Briefings in Bioinformatics**, 2024.

SKILLS

Engineering: Python, PyTorch, PyTorch Lightning, Numpy, Hydra, Weights & Biases, Hugging Face, Slurm, Git, Linux

Research: Generative modeling (flow matching, diffusion), protein language modeling, protein design workflows (RFDiffusion, ProteinMPNN, AlphaFold / Boltz)

PRESENTATIONS

Poster: *forge: sequence-based binder design with latent flow matching.* **Machine Learning in Structural Biology (MLSB)**, San Diego, USA, December 2025.

Poster and Lightning Talk: *Advancing the embedding transfer paradigm: benchmarking text-integrated protein language models and embedding fusion.* **RECOMB**, Seoul, South Korea, April 2025.

Poster: *Benchmarking text-integrated protein language model embeddings and embedding fusion on diverse downstream tasks.* **Machine Learning in Structural Biology (MLSB) Workshop at NeurIPS**, Vancouver, Canada, December 2024.

AWARDS

2024-2025 Distinguished Graduate Student Fellowship, University of California, San Diego 2025

- Awarded for research excellence; includes full fellowship support for one quarter and a \$3,000 research stipend.

The John Stauffer Scholarship for Academic Merit, Pomona College 2022

- \$12,000 scholarship awarded to one senior annually in the natural sciences.

The Frank Parkhurst Brackett, Jr., and Davida Wark Brackett Prize, Pomona College 2022

- Awarded to a student as an incentive to excellence in advancing the study of chemistry.

Phi Beta Kappa Award, Pomona College 2022

- \$2,750 award endowed by the Pomona Chapter of Phi Beta Kappa.

REVIEWING

Genome Biology 2026

BMC Genomics 2026

Machine Learning for Structural Biology (MLSB) 2025

NeurIPS Workshop on Machine Learning for Structural Biology 2024

OUP Bioinformatics 2024